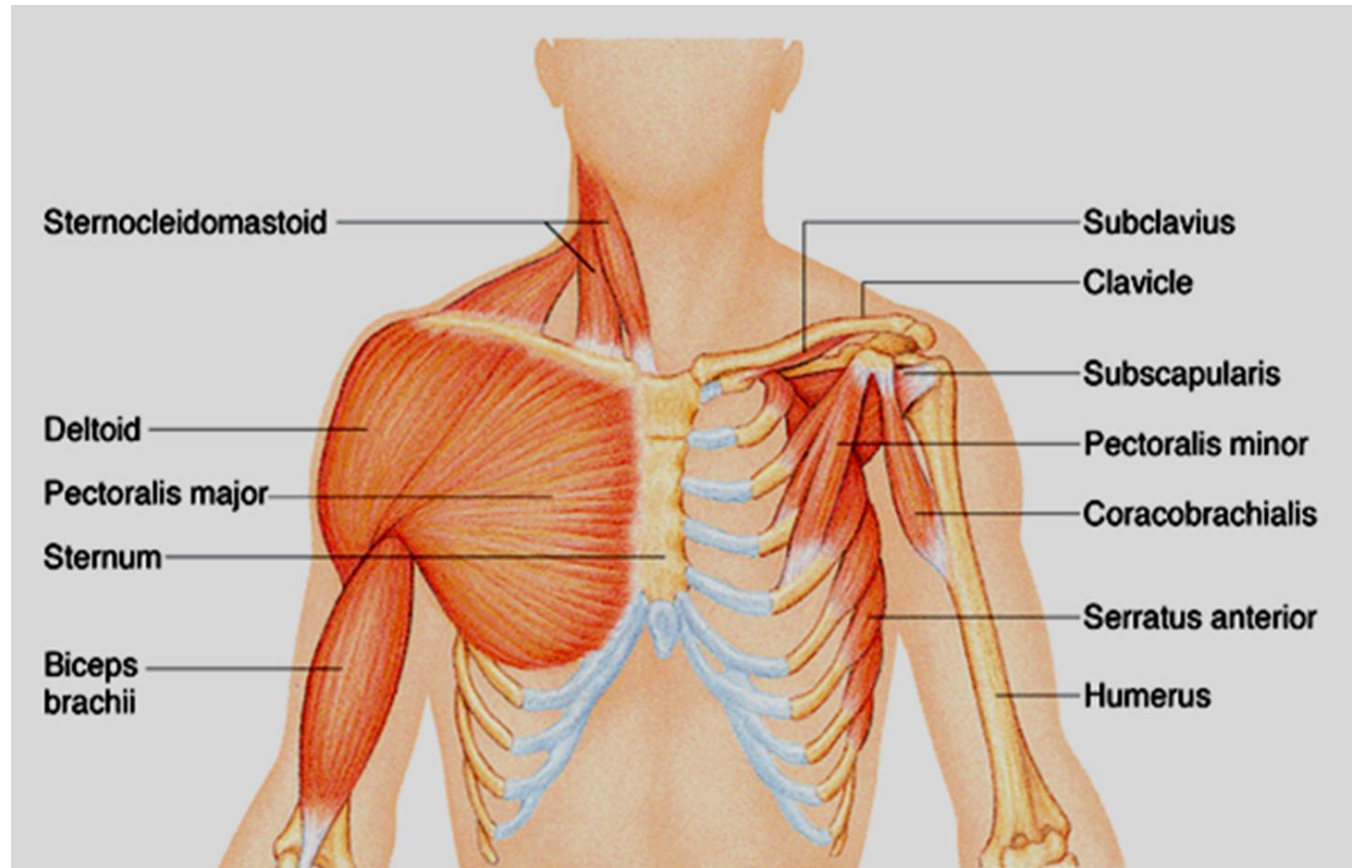


LECTURE FOUR

PECTORAL, SUPERFICIAL BACK AND
SHOULDER REGIONS

Anterior Axioappendicular Muscles

- Four muscles that move the pectoral girdle
 - Pectoralis major
 - Pectoralis minor
 - Subclavius
 - Serratus anterior



Pectoralis major

- Fan-shaped
- Covers superior part of thorax
- Lateral border forms anterior wall of axilla (anterior axillary fold)
- With deltoid form deltopectoral groove
- Superiorly separate with deltoid and with clavicle form deltopectoral triangle

❑ Origin

- Clavicular head: medial half of clavicle
- Sternocostal head: sternum, superior six costal cartilages, aponeurosis of external oblique muscle

❑ Insertion

- Intertubercular groove of humerus

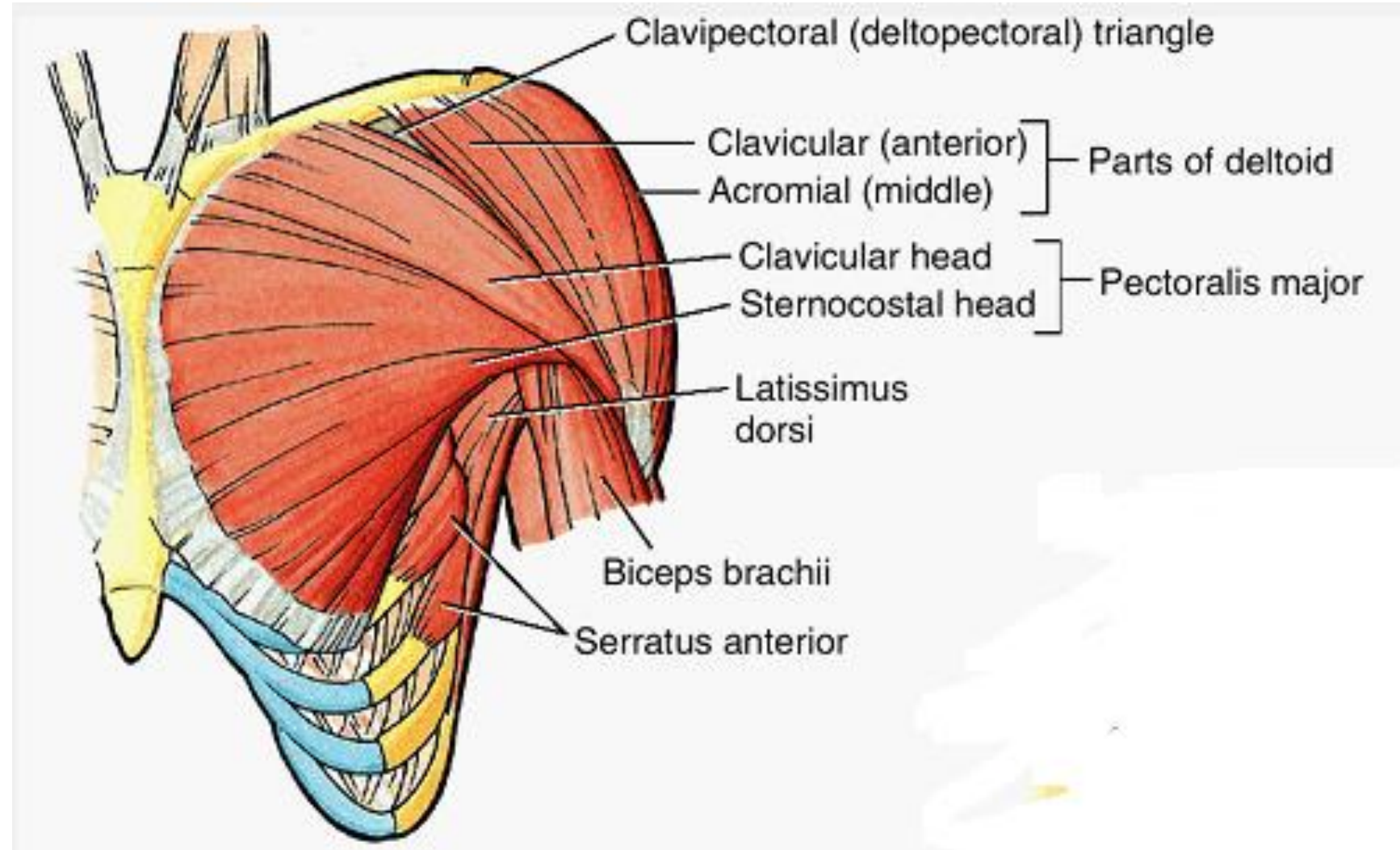
❑ Innervation

- Lateral and medial pectoral nerves

Cont....

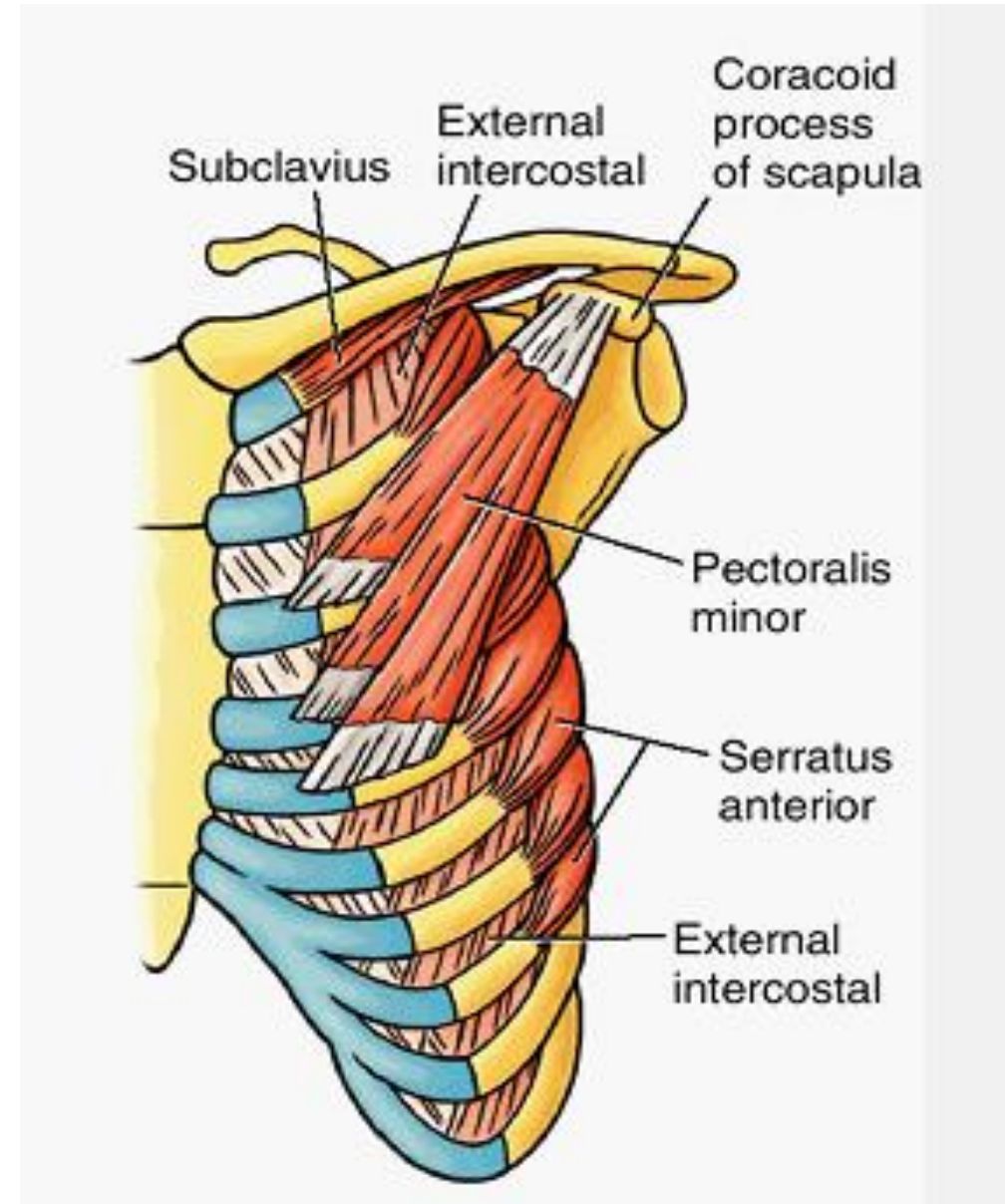
□ Action

- **Adducts and medially rotates** humerus; draws scapula anteriorly and inferiorly
- Acting alone, **clavicular head flexes** humerus and **sternocostal head extends** it from the flexed position



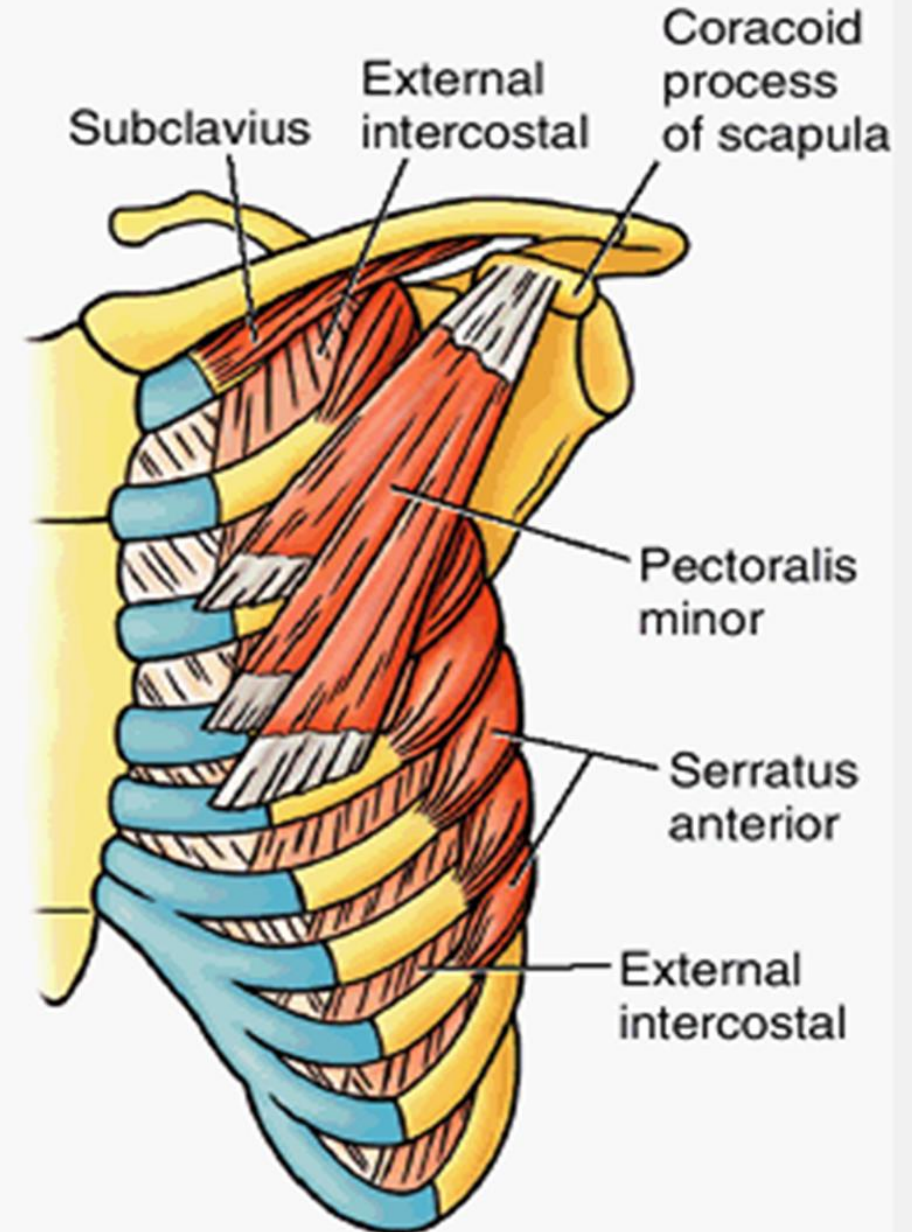
Pectoralis minor

- Triangular
- Lies in anterior wall of axilla
- Covered by pectoralis major
- ❑ Origin: 3rd to 5th ribs near their costal cartilages
- ❑ Insertion: coracoid process of scapula
- ❑ Innervation: **Medial pectoral nerve**
- ❑ Action:
 - Stabilizes scapula by drawing it inferiorly and anteriorly against thoracic wall
 - Elevates ribs in forced inspiration



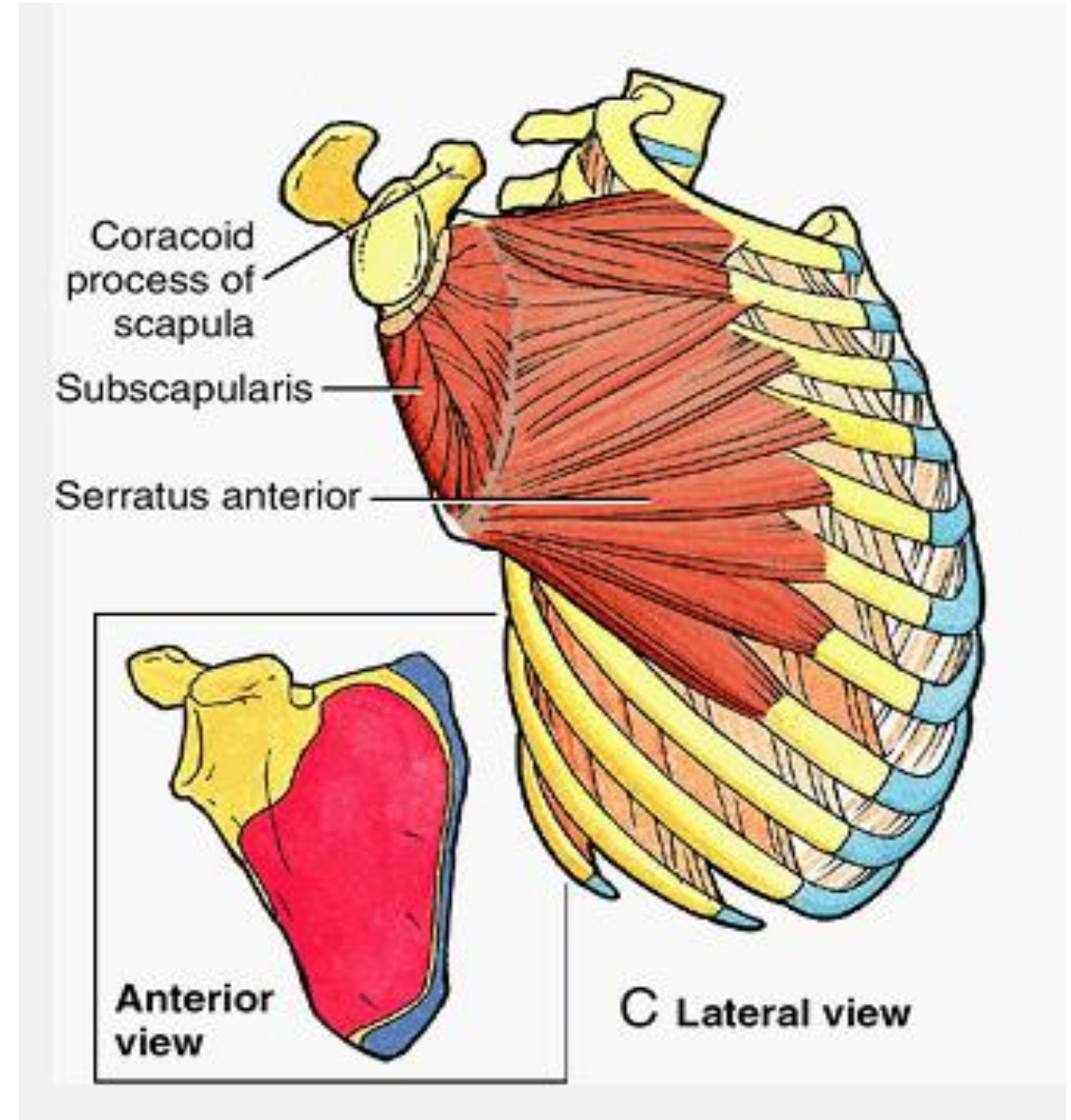
Subclavius

- Small, round that lies horizontally inferior to clavicle
- Protect subclavian vessels and superior trunk of brachial plexus
- Origin: Junction of 1st rib and its costal cartilage
- Insertion: Inferior surface of middle third of clavicle
- Innervation: **Nerve to subclavius**
- Action: Anchors and depresses clavicle



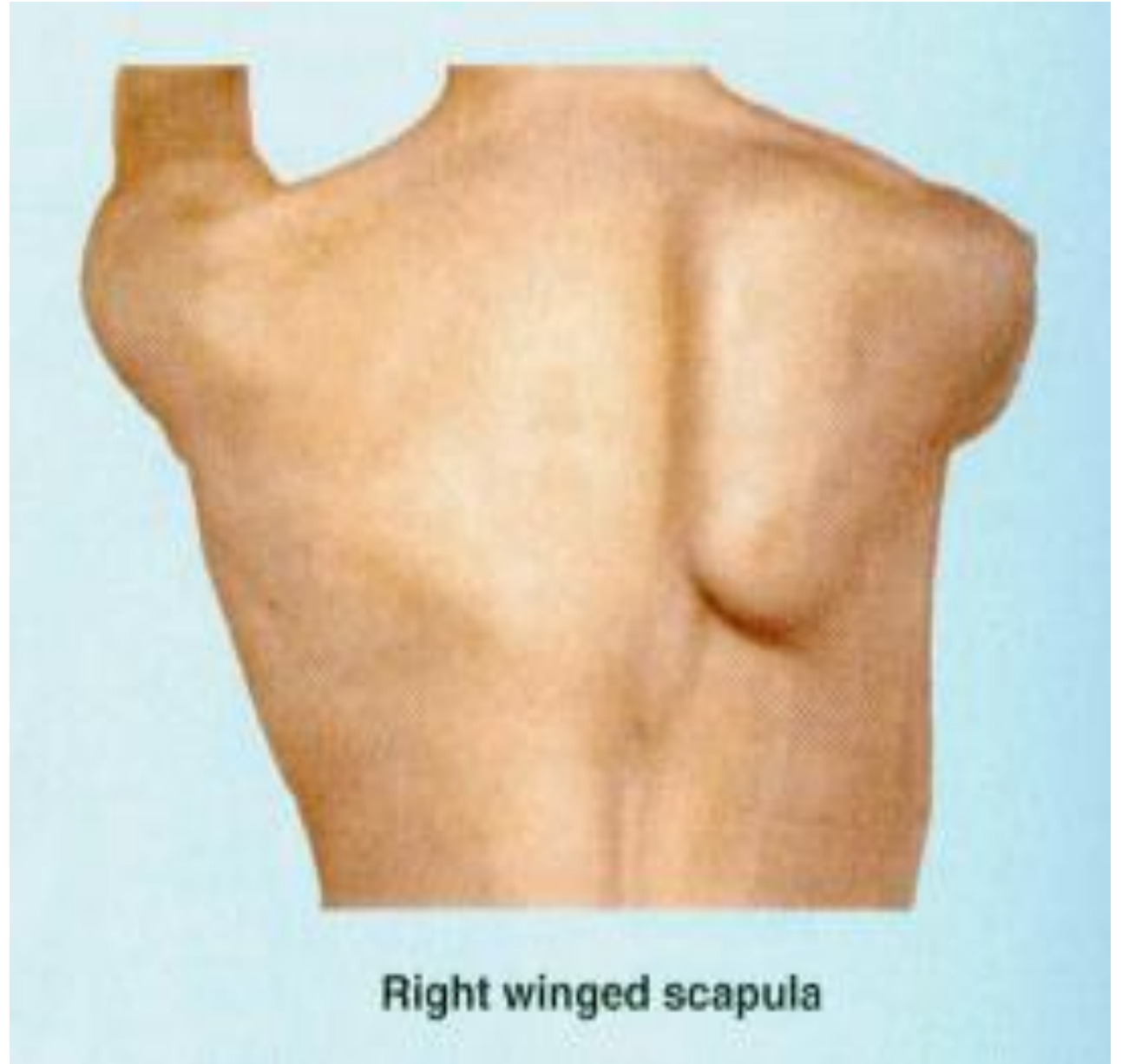
Serratus anterior

- Overlies lateral part of thorax
- Forms medial wall of axilla
- Anchors scapula, enabling other muscles to use it as a fixed bone for movement of humerus
- **Origin:** External surfaces of lateral parts of 1st to 8th ribs
- **Insertion:** Anterior surface of medial border of scapula
- **Innervation:** Long thoracic nerve
- **Action:** Protracts scapula and holds it against thoracic wall; rotates scapula

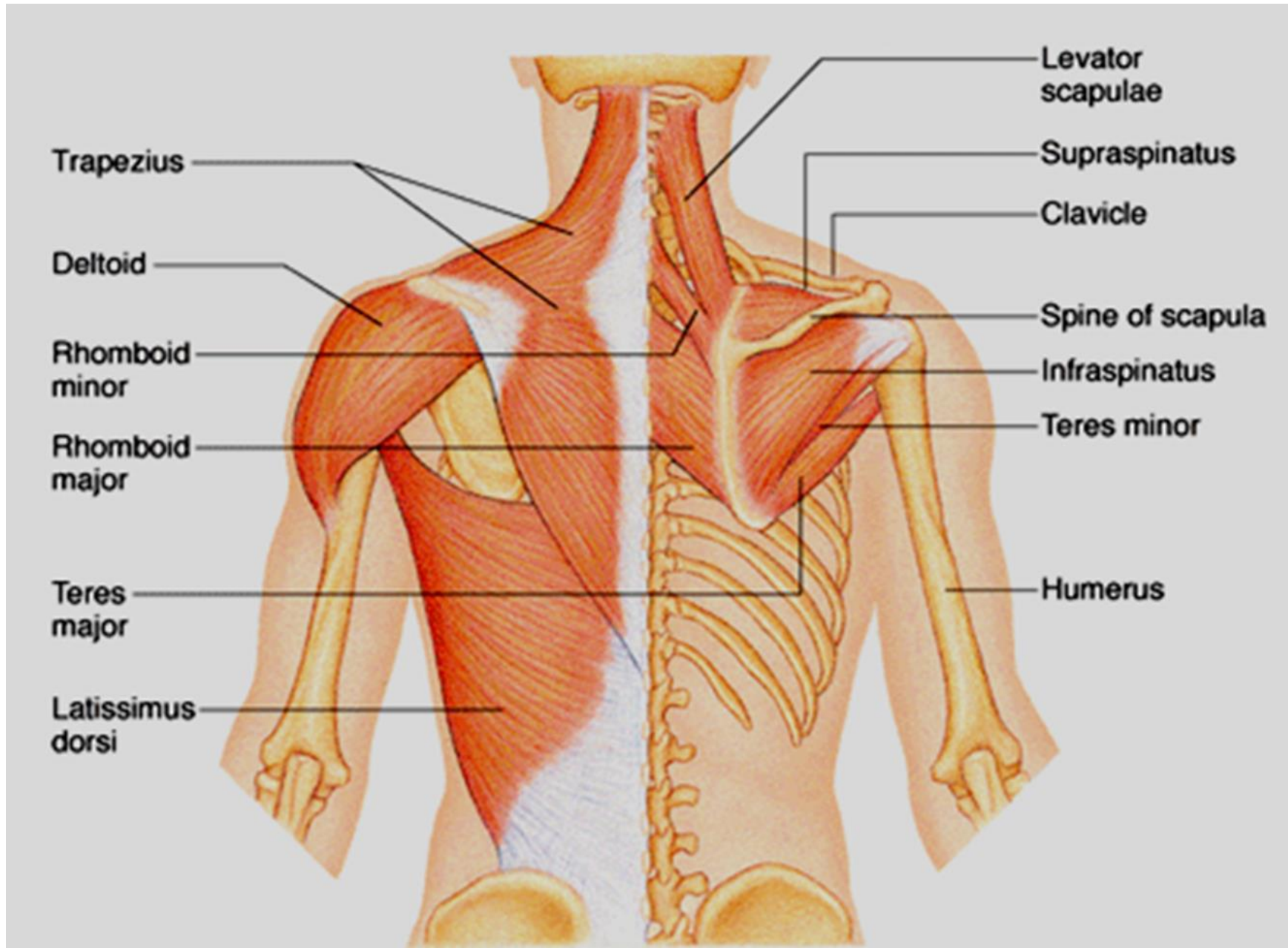


Paralysis of serratus anterior

- Due to injury of long thoracic nerve
- Medial border of the scapula moves laterally and posteriorly away from thoracic wall
 - **winged scapula**
- Arm cannot be abducted above horizontal position

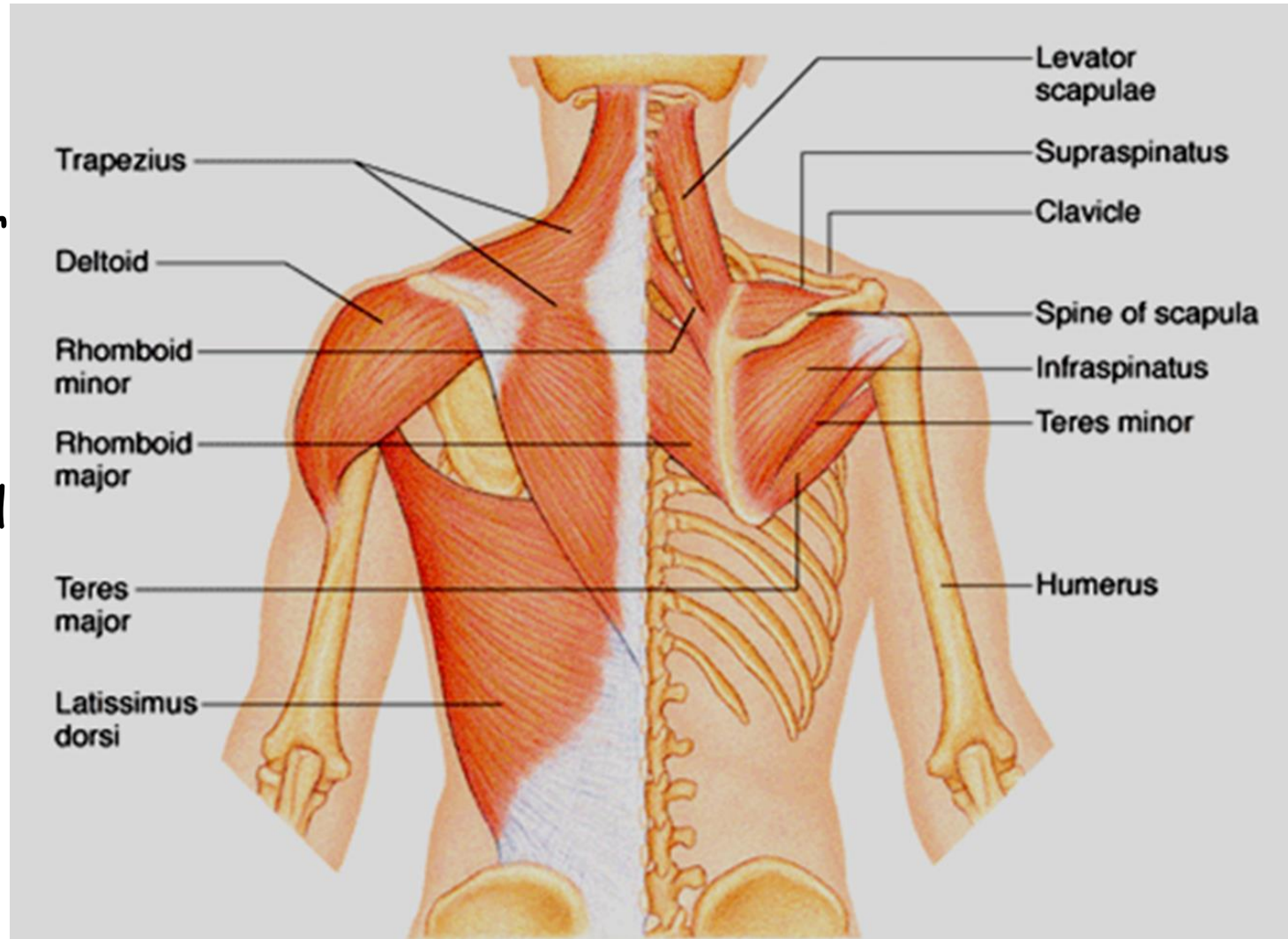


Posterior Axioappendicular Muscles



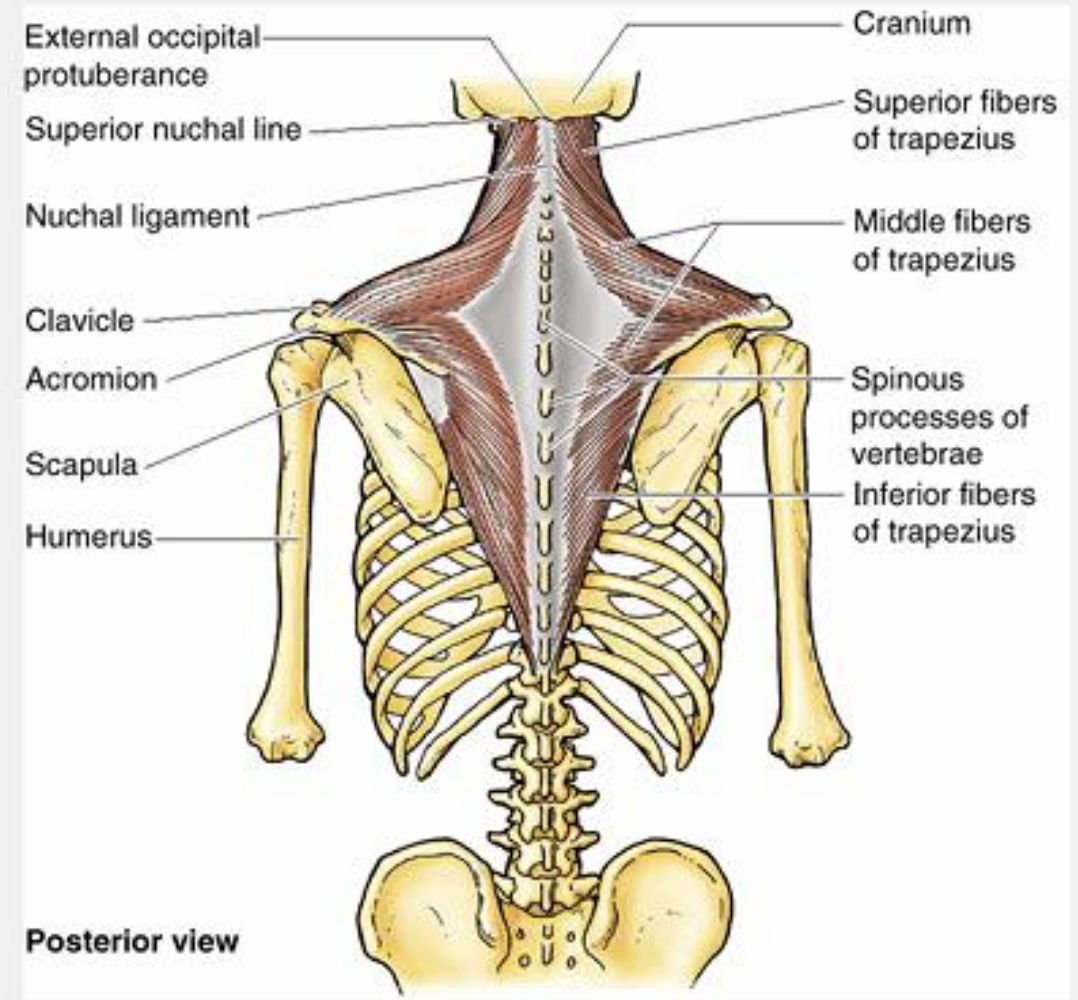
Posterior Axioappendicular Muscles

- Attach the superior appendicular skeleton to the axial skeleton
- Divided into three groups
 - **Superficial** (extrinsic shoulder muscles : **trapezius** and **latissimus dorsi**)
 - **Deep** (extrinsic shoulder) muscles : **levator scapulae** and **rhomboids**
 - **Scapulohumeral** (intrinsic shoulder) muscles: **deltoid**, **teres major**, and the four rotator cuff muscles (**supraspinatus**, **infraspinatus**, **teres minor**, and **subscapularis**)



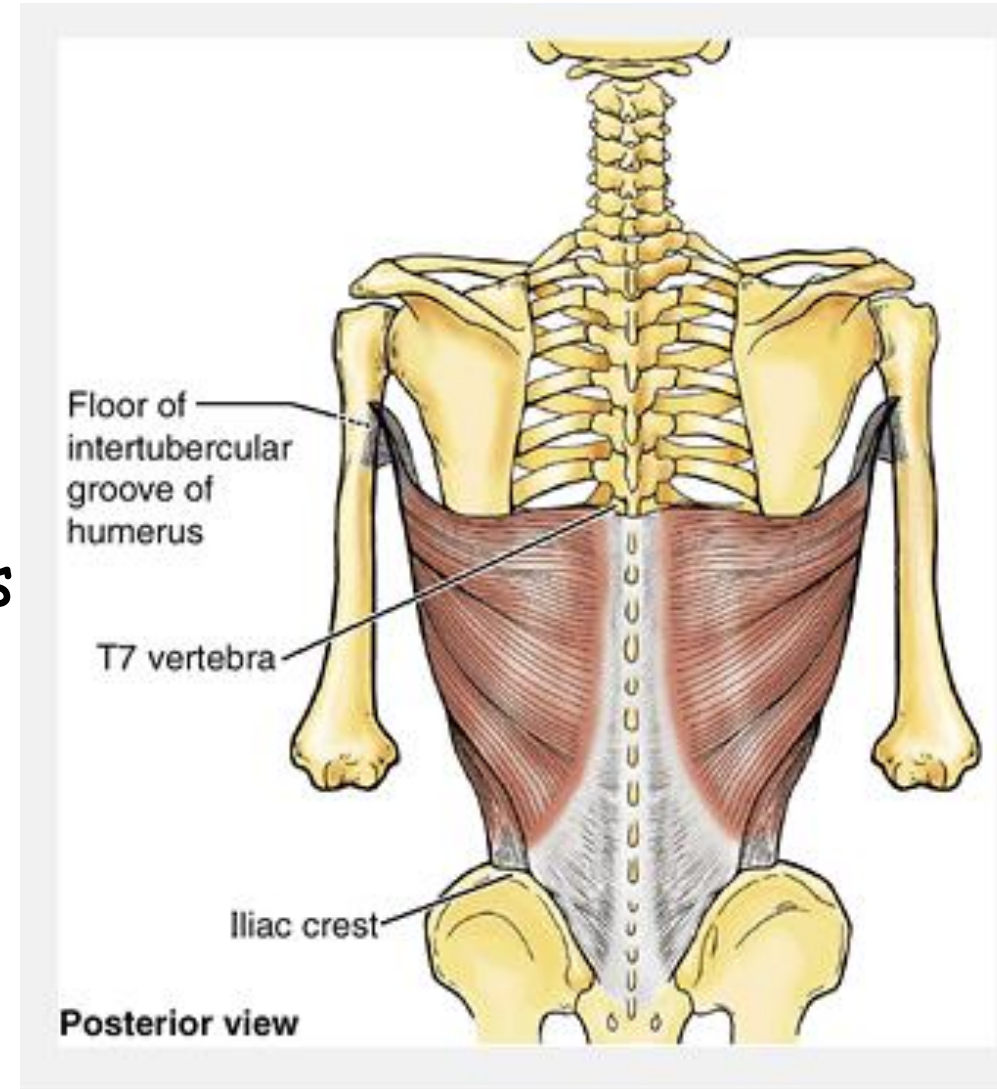
Trapezius

- **Origin:** Medial third of superior nuchal line; external occipital protuberance; nuchal ligament; spinous processes of C7 to T12 vertebrae
- **Insertion:** Lateral third of clavicle; acromion and spine of scapula
- **Innervation:** Accessory nerve (CN XI) (motor fibers) and C3, C4 spinal nerves (pain and proprioceptive fibers)
- **Action:**
 - descending part elevates scapula
 - ascending part depresses scapula
 - middle part (or all parts together) retracts scapula
 - descending and ascending parts act together to rotate glenoid cavity superiorly



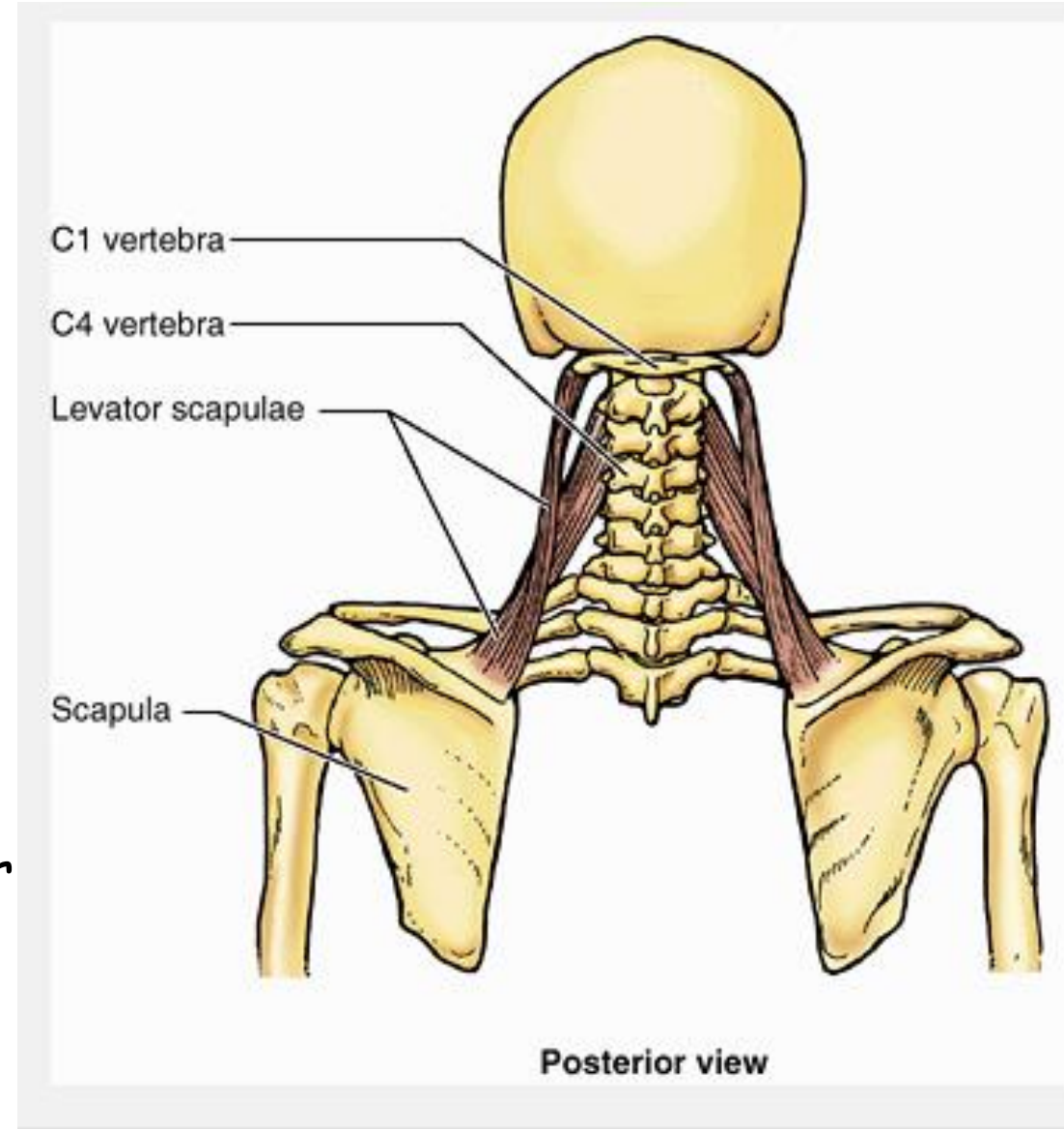
Latissimus dorsi

- Fan-shaped
- Covers wide area of back
- Passes from trunk to humerus
- Raises trunk to the arm
- **Origin:** Spinous processes of inferior 6 thoracic vertebrae, thoracolumbar fascia, iliac crest, and inferior 3 or 4 ribs
- **Insertion:** intertubercular groove of humerus
- **Innervation:** Thoracodorsal nerve
- **Action:** Extends, adducts, and medially rotates humerus; raises body toward arms during climbing



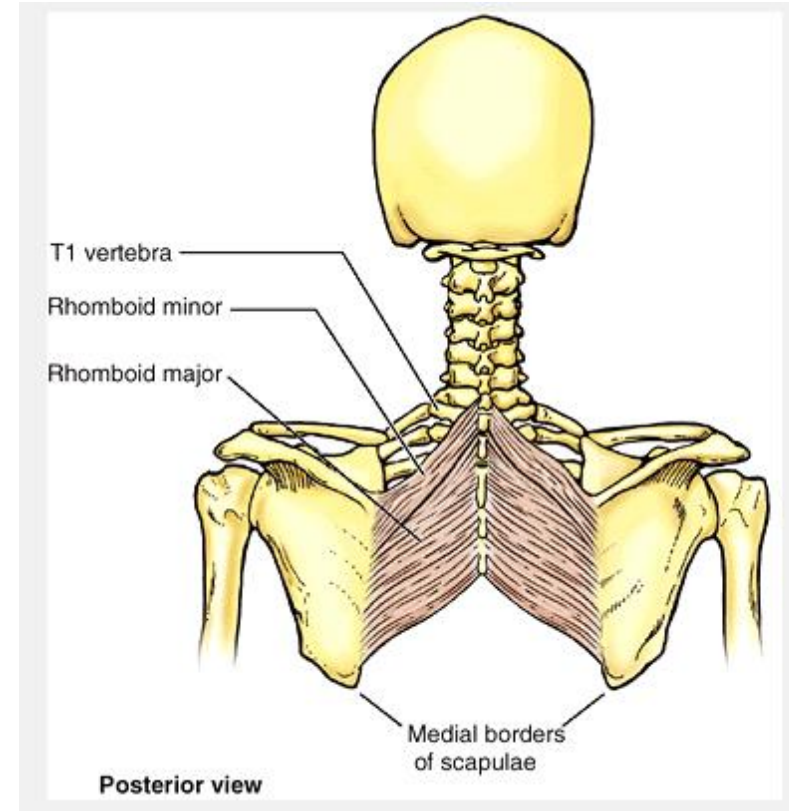
Levator scapulae

- Lies deep to SCM and trapezius
- Origin: Transverse processes of C1 to C4 vertebrae
- Insertion: Medial border of scapula superior to root of spine
- Innervation: Dorsal scapular (C5) and cervical (C3, C4) nerves
- Action:
 - Elevates scapula and tilts its glenoid cavity inferiorly by rotating scapula with the rhomboid and pectoralis minor
 - Acting bilaterally, extend the neck
 - Acting unilaterally, contribute to lateral flexion of the neck



Rhomboids (major and minor)

- Lie deep to trapezius
- Form parallel bands that pass inferolaterally from vertebrae to medial border of the scapula
- Origin:
 - Minor: nuchal ligament; spinous processes of C7 and T1 vertebrae
 - Major: spinous processes of T2 to T5 vertebrae
- Insertion:
 - Minor: medial end of scapular spine
 - Major: medial border of scapula from level of spine to inferior angle
- Innervation: Dorsal scapular nerve
- Action:
 - Retract scapula and rotate it to depress glenoid cavity
 - fix scapula to thoracic wall during movements of upper limb

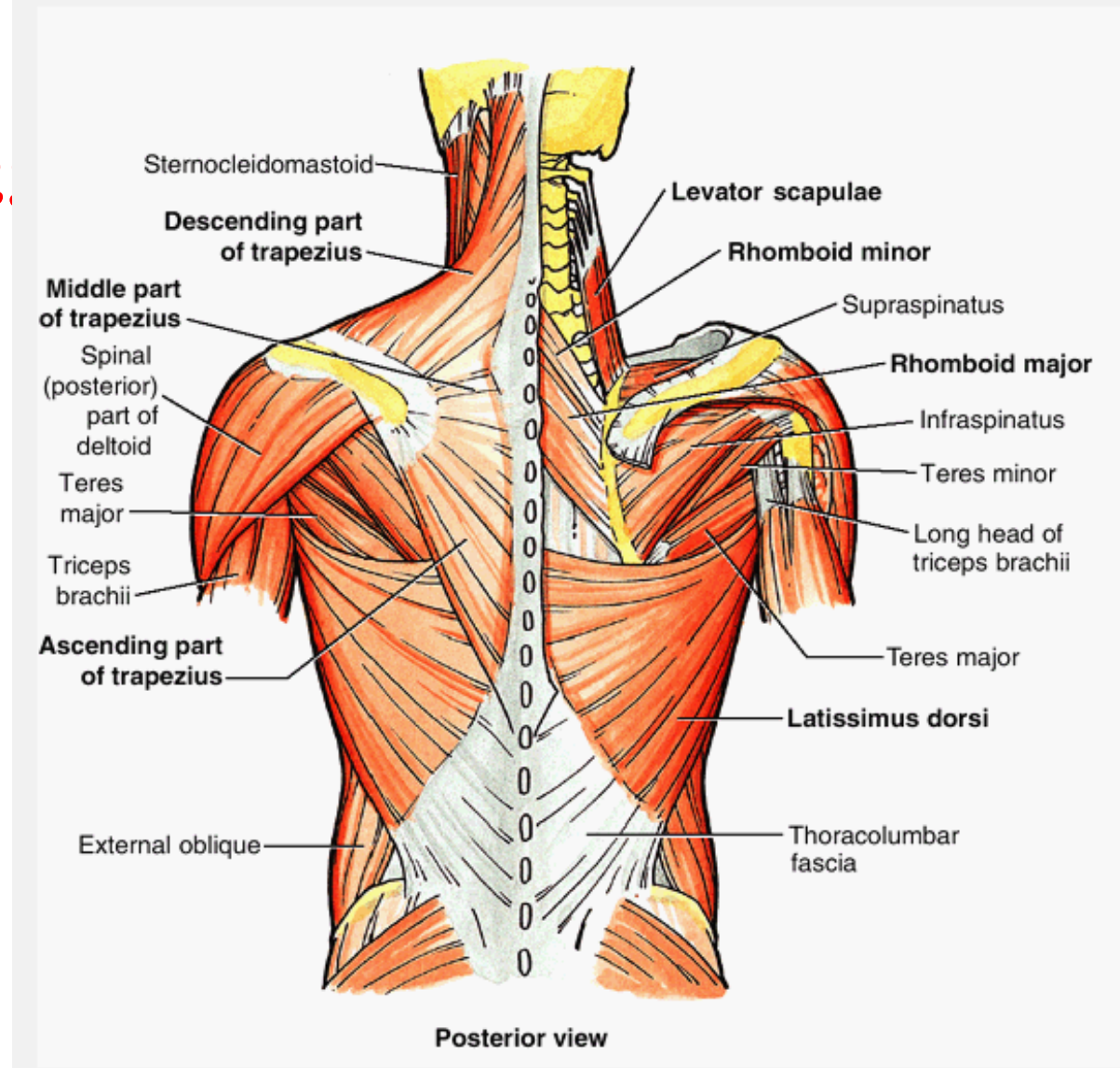


Scapulohumeral muscle.

- Six muscles

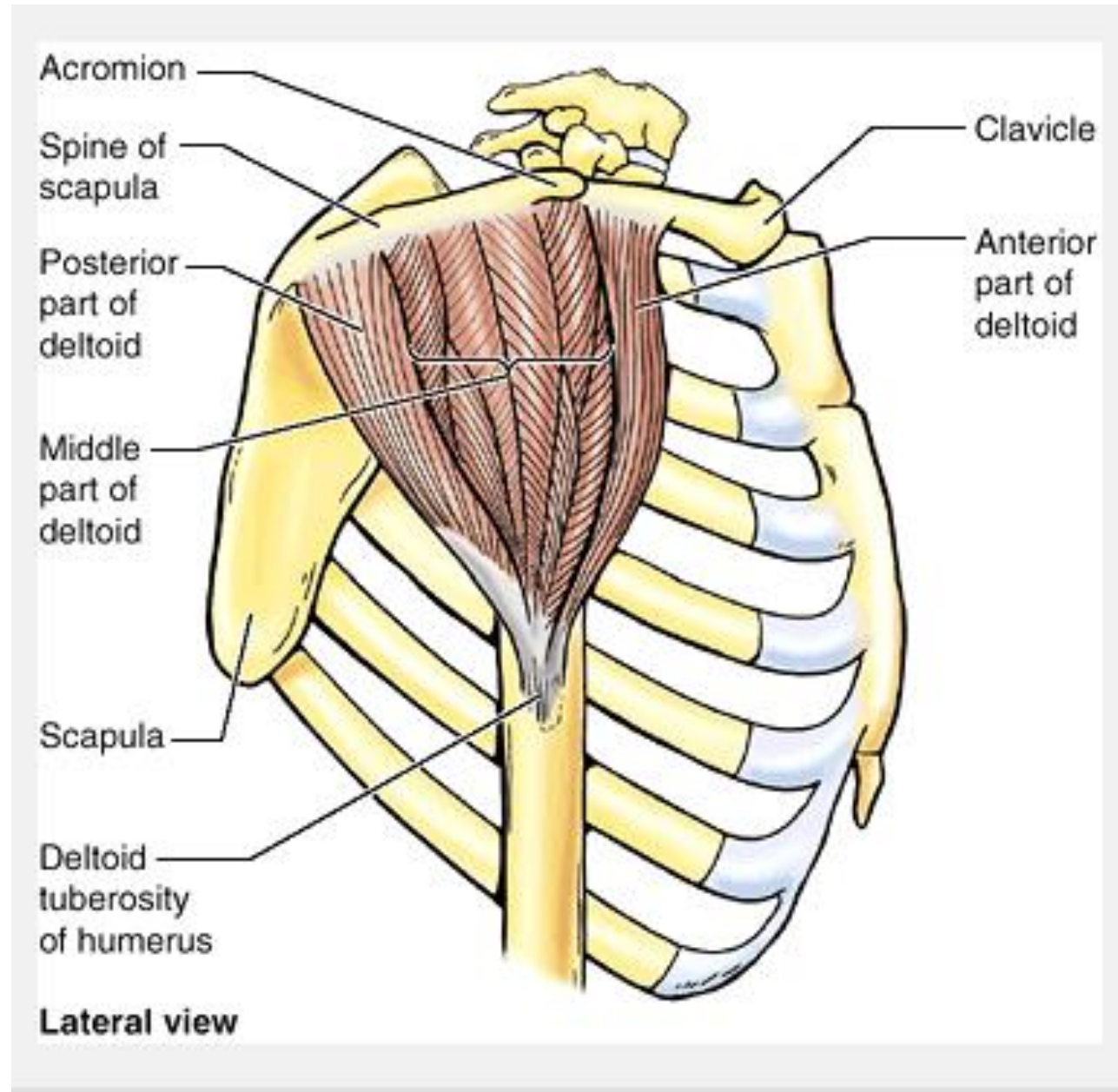
- Deltoid
- Teres major
- Supraspinatus
- Infraspinatus
- Subscapularis
- Teres minor

- Short muscles that pass from scapula to humerus
- Act on glenohumeral joint



Deltoid muscle

- **Origin:** Lateral third of clavicle; acromion and spine of scapula
- **Insertion:** Deltoid tuberosity of humerus
- **Innervation:** Axillary nerve
- **Action:**
 - Anterior part: flexes and medially rotates arm
 - Middle part: abducts arm
 - Posterior part: extends and laterally rotates arm



Teres major

- Thick rounded muscle that lies on the inferolateral third of the scapula
- Along with the deltoid and rotator cuff muscles it is an important stabilizer of the humeral head in the glenoid cavity during movement
- **Origin:** Posterior surface of inferior angle of scapula
- **Insertion:** intertubercular groove of humerus
- **Innervation:** Lower subscapular nerve
- **Action:** Adducts and medially rotates arm

Rotator cuff muscles

- Four of the scapulohumeral muscles: **supraspinatus**, **infraspinatus**, **teres minor**, and **subscapularis**
- Called rotator cuff because they form a **musculotendinous cuff** around glenohumeral joint
- All except the **supraspinatus** are rotators of the humerus
- The supraspinatus initiates and assists the deltoid in the abduction of the arm
- The tendons of the rotator cuff muscles blend with the joint capsule of the glenohumeral joint, which protects the joint and gives it stability
- Tonic contraction of these muscles holds the relatively large head of the humerus firmly against the small and shallow glenoid cavity during arm movements

Supraspinatus

- Origin: Supraspinous fossa of scapula
- Insertion: greater tubercle of humerus
- Innervation: Suprascapular nerve
- Action: Initiates and assists deltoid in abduction of arm and acts with rotator cuff muscles

Infraspinatus

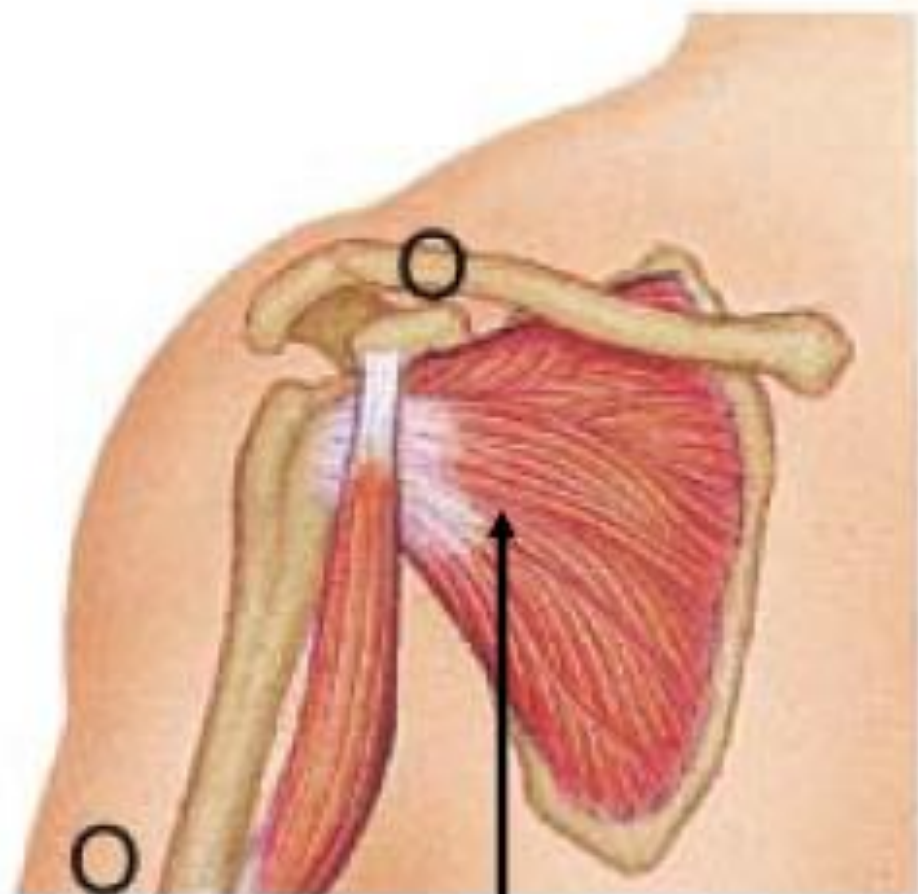
- Origin: Infraspinous fossa of scapula
- Insertion: greater tubercle of humerus
- Innervation: Suprascapular nerve
- Action: Laterally rotate arm; help hold humeral head in glenoid cavity of scapula

Teres minor

- Origin: Middle part of lateral border of scapula
- Insertion: greater tubercle of humerus
- Innervation: Axillary nerve
- Action: Laterally rotate arm; help hold humeral head in glenoid cavity of scapula

Subscapularis

- Origin: Subscapular fossa
- Insertion: Lesser tubercle of humerus
- Innervation: Upper and lower subscapular nerves
- Action: Medially rotates and adduct arm; helps hold humeral head in glenoid cavity



● Subscapularis



Abductor

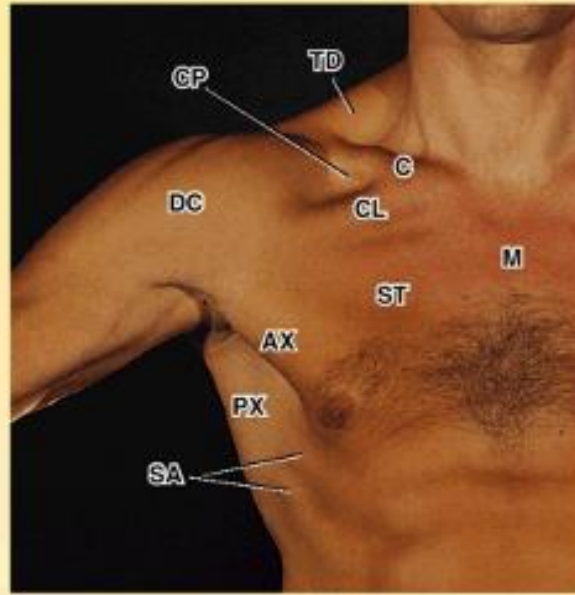
Supraspinatus ●

Infraspinatus ●

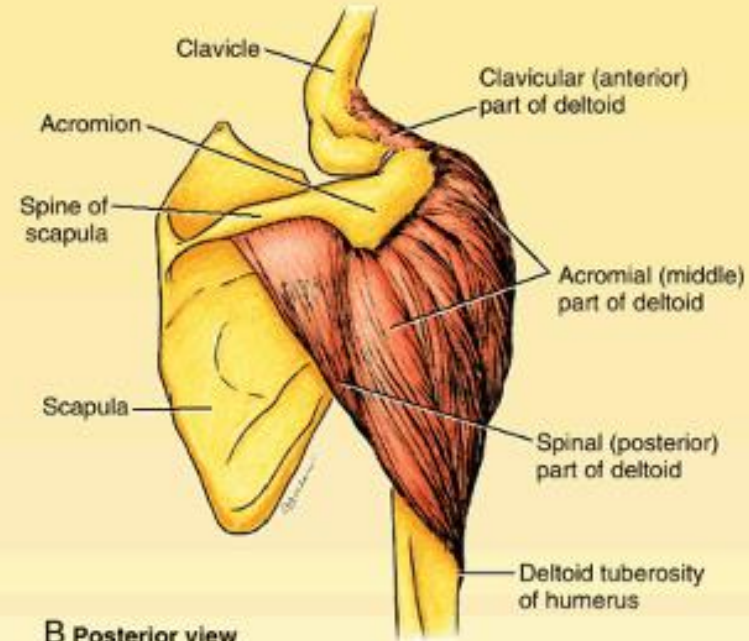
Teres minor ●

Adductors

Surface anatomy

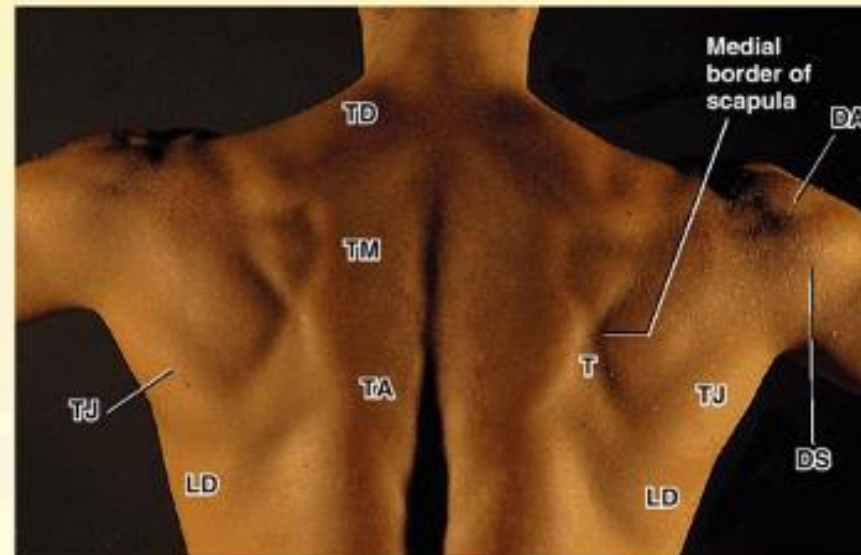


A Anterior view



B Posterior view

Key	
AX	Anterior axillary fold
C	Clavicle
CL	Clavicular head of pectoralis major
CP	Clavipectoral triangle
DA	Acromial (middle) part of deltoid
DC	Clavicular (anterior) part of deltoid
DS	Spinal (posterior) part of deltoid
LD	Latissimus dorsi
M	Manubrium
PX	Posterior axillary fold
SA	Serratus anterior
ST	Sternocostal head of pectoralis major
T	Triangle of auscultation
TA	Ascending part of trapezius
TD	Descending part of trapezius
TJ	Teres major
TM	Middle part of trapezius



C Posterior view